Midterm Exam Solution
Electromagnetic Theory
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50 Points

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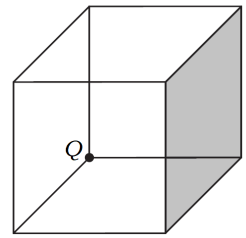
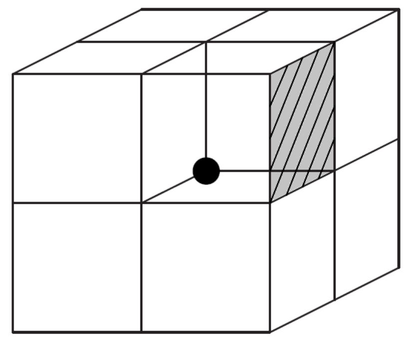
Q1. (10 points) A **point charge** Q sits at a corner of a cube, as shown here. What is the flux of $\boldsymbol{E}$ through the shaded side?



Hint: The position of charge may be considered to be at the origin. To evaluate flux in *Coulomb*, you need to solve the following equation over the specified shaded surface:

$$\psi = \epsilon_0 \int \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{S}$$

Solution: Think of this cube as one of 8 surrounding the charge. Each of the 24 squares which make up the surface of this larger cube gets the same flux as every other one, so:



$$\epsilon_0 \int_{\substack{\text{one}\\ \text{face}}} \boldsymbol{E} \cdot d\boldsymbol{S} = \frac{\epsilon_0}{24} \int_{\substack{\text{whole}\\ \text{large}\\ \text{cube}}} \boldsymbol{E} \cdot d\boldsymbol{S} = \frac{\epsilon_0}{24} \oint \boldsymbol{E} \cdot d\boldsymbol{S} = \frac{\epsilon_0}{24} \frac{Q}{\epsilon_0} = \frac{Q}{24}$$

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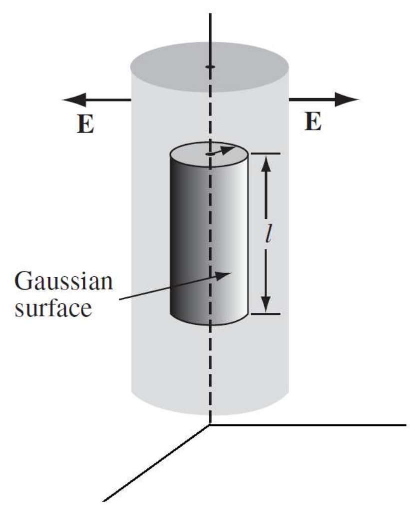
Q2. (10 points) A **long solid cylinder** (of radius a) carries a volumetric charge density that is proportional to the distance from the axis: $\rho_v = k\rho$, for some constant k. Using Gauss's theorem, show that the electric field $\boldsymbol{E}$ inside this cylinder is given as:

$$\boldsymbol{E} = \frac{k}{3\epsilon_0} \rho^2 \mathbf{a}_\rho$$



Hint: It is an infinite cylinder, for which $\boldsymbol{E} = E_\rho \mathbf{a}_\rho$ is true. You may consider a Gauss's closed surface (an inner cylinder of height l and field radius ρ, as shown here) and evaluate the following:

$$\oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{S} = \frac{1}{\epsilon_0} Q_{\text{enc.}}$$

Solution: For R.H.S., we introduce an auxiliary variable such that $0 \leq \check{\rho} \leq \rho \leq a$.

$$\text{L.H.S.} = \oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{S} = \int_{\substack{\text{outer}\\ \text{cylindrical}\\ \text{surface}}} \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{S} = \int_{z=0}^{L} \int_{\phi=0}^{2\pi} \left(E_\rho \mathbf{a}_\rho\right) \cdot \left(\rho d\phi dz \mathbf{a}_\rho\right) = 2\pi L \rho E_\rho$$

$$\text{R.H.S.} = \frac{Q_{\text{enc.}}}{\epsilon_0} = \frac{1}{\epsilon_0} \int \rho_v dv = \frac{k}{\epsilon_0} \iiint \check{\rho}\ \check{\rho} d\check{\rho} d\phi dz = \frac{k}{\epsilon_0} \int_{z=0}^{L} \int_{\phi=0}^{2\pi} \int_{\check{\rho}=0}^{\rho} \check{\rho}^2 d\check{\rho} d\phi dz = \frac{k}{\epsilon_0} \frac{\rho^3}{3} 2\pi L$$

Comparing the two sides, we get the required equation.

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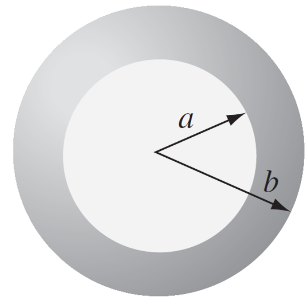
Q3. (20 points) A **thick spherical shell** carries charge density

$$\rho_v = \begin{cases} \dfrac{k}{r^2}, & a \le r \le b \\ 0, & \text{otherwise.} \end{cases}$$



(a) Find the electric field $\boldsymbol{E}$ in the three regions:
. **(i)** $r < a$, **(ii)** $a < r < b$, **(iii)** $r > b$.

(b) Plot $|\boldsymbol{E}|$ as a function of r, for the case $b = 2a$.

Hint: Owing to symmetry, $\boldsymbol{E} = E_r \mathbf{a}_r$ is true.

Solution: For $r < 0$, $\boldsymbol{E} = 0$. For $a < r < b$, repeat in a similar fashion as I have shown in Q2; introduce an auxiliary variable such that $a \le \check{r} \le r \le b$.

$$\text{L.H.S.} = \oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{S} = \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \big(E_r \mathbf{a}_r\big) \cdot \big(r^2 \sin(\theta)\, d\phi\, d\theta\, \mathbf{a}_r\big) = 4\pi r^2 E_r$$

$$\text{R.H.S.} = \frac{Q_{\text{enc.}}}{\epsilon_0} = \frac{1}{\epsilon_0} \int \rho_v dv = \frac{k}{\epsilon_0} \iiint \frac{1}{\check{r}^2}\, \check{r}^2 \sin(\theta)\, d\check{r}\, d\phi\, d\theta = \frac{k}{\epsilon_0} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \int_{\check{r}=a}^{r} \sin(\theta)\, d\check{r}\, d\phi\, d\theta = \frac{k}{\epsilon_0} 4\pi(r-a)$$

Comparing the two, we obtain:

$$\boldsymbol{E} = E_r \mathbf{a}_r = \frac{k}{\epsilon_0} \frac{(r-a)}{r^2} \mathbf{a}_r, \quad a \le r \le b$$

For $r > b$, we solve as follows (we do not need to introduce any auxiliary variable in the evaluation of R.H.S.):

$$\text{L.H.S.} = \oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{S} = \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \big(E_r \mathbf{a}_r\big) \cdot \big(r^2 \sin(\theta)\, d\phi\, d\theta\, \mathbf{a}_r\big) = 4\pi r^2 E_r \quad \text{(same as above)}$$

$$\text{R.H.S.} = \frac{Q_{\text{enc.}}}{\epsilon_0} = \frac{1}{\epsilon_0} \int \rho_v dv = \frac{k}{\epsilon_0} \iiint \frac{1}{r^2}\, r^2 \sin(\theta)\, dr\, d\phi\, d\theta = \frac{k}{\epsilon_0} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \int_{r=a}^{b} \sin(\theta)\, dr\, d\phi\, d\theta = \frac{k}{\epsilon_0} 4\pi(b-a)$$

Comparing the two, we obtain:

$$\boldsymbol{E} = E_r \mathbf{a}_r = \frac{k}{\epsilon_0} \frac{(b-a)}{r^2} \mathbf{a}_r, \quad b \le r < \infty$$

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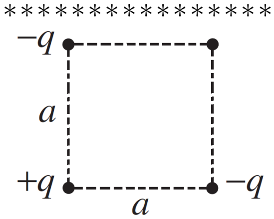
Q4. (10 points)



(a) Three charges are situated at the corners of a square (each side is length a). How much work does it take to bring in another charge, $+q$, from far away and place it in the fourth corner?

(b) How much work does it take to assemble the whole configuration of four charges?

Solution: Do it yourself to show (a) $W_4 = \frac{q^2}{4\pi\epsilon_0 a}\left(\frac{1}{\sqrt{2}} - 2\right)$, and (b) $W_{\text{total}} = \frac{2q^2}{4\pi\epsilon_0 a}\left(\frac{1}{\sqrt{2}} - 2\right)$.

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Q5. (10 points) Find the energy of a uniformly charged spherical (hollow) shell of total charge q and radius a.

Hint: The sphere acts like a point charge; if the total charge is q, then the field $\boldsymbol{E}$ outside the sphere is given as

$$\boldsymbol{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\mathbf{a}_r \quad (a \leqslant r < \infty)$$

Use $W_E = (\epsilon_0/2)\int E^2\,\mathrm{d}v$, where the volume (triple) integral is evaluated for the **entire space**; say we integrate (in the spherical coordinate system) from the surface of the sphere to infinity. We can exclude sphere's inner region because the field is zero inside.

Solution: Do it yourself to show:

$$W_E = \frac{\epsilon_0}{2}\int E^2\,dv = \frac{q^2}{8\pi\epsilon_0 a}$$

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Q6. (10 points) Determine the total charge

(a) On the cylinder $\rho = 4$, $0 < z < 10$, if $\rho_s = \rho z^3$ [nC/m^2]

(b) Within the cylinder $r = 3$, if $\rho_v = \frac{10}{r^2 \sin\theta}$ [C/m^3]

(c) On line $0 < x < 5$, if $\rho_l = 12x^3$ [C/m]

Solution: This is easy; do it yourself, and talk to each other to get your answers verified.

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